The spectral problem for the Kronig–Penney model of the Dirac operators in terms of Schur's algorithm

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Dirac operator on the half-line

The Dirac operator $\mathcal{D}_q$ on $\mathbb{R}_+$ with a square summable potential $q \in L^2(\mathbb{R}_+)$ is defined by the differential expression

$$\mathcal{D}_q : X \mapsto JX' + QX, \qquad Q = \begin{pmatrix} \operatorname{Im} q & \operatorname{Re} q \\ \operatorname{Re} q & -\operatorname{Im} q \end{pmatrix}, \qquad J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

The operator $\mathcal{D}_q$ defined on the domain

$$\{X \colon X \in \mathrm{AC}_{\mathrm{loc}} \cap L^2, \ \ \langle X(0), \left(\begin{smallmatrix} \sin\alpha \\ \cos\alpha \end{smallmatrix}\right) \rangle_{\mathbb{C}^2} = 0, \ \ \mathcal{D}_q X \in L^2\}$$

is the densely defined self-adjoint operator on $L^2(\mathbb{R}_+, \mathbb{C}^2)$.

Weyl function of Dirac operator

The generalized eigenfunction equation

$$\mathcal{D}_q f = \lambda f, \qquad\qquad Jf' + Qf = \lambda f.$$

The fundamental solution $N = \left(\begin{smallmatrix} n_{11} & n_{12} \\ n_{21} & n_{22} \end{smallmatrix}\right)$ satisfies

$$JN'(x,z) + Q(x)N(x,z) = zN(x,z), \quad N(0,z) = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}\right), \quad z \in \mathbb{C}.$$

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The Weyl function $m_q \colon \mathbb{C}_+ \to \mathbb{C}_+$ of $\mathcal{D}_q$ is defined by

$$m_q(z) = \lim_{x \to +\infty} \frac{n_{22}(x,z)}{n_{21}(x,z)}, \qquad z \in \mathbb{C}_+.$$

The Schur function $f_q \colon \mathbb{C}_+ \to \mathbb{D}$ is given by

$$f_q = \frac{m_q - i}{m_q + i}.$$

Canonical Hamiltonian systems

Canonical Hamiltonian system on $\mathbb{R}_+$ is the differential equation

$$JX'(x,z) = z\mathcal{H}(x)X(x,z), \qquad x \in \mathbb{R}_+, \quad z \in \mathbb{C},$$

where $\mathcal{H}$ is a Hamiltonian,

$$\mathcal{H} : x \mapsto \begin{pmatrix} h_1 & h \\ h & h_2 \end{pmatrix}, \quad h, h_1, h_2 \in L^1_{\mathrm{loc}}(\mathbb{R}_+), \quad \mathcal{H}(x) \geqslant 0.$$

The fundamental solution M solves

$$JM'(x,z) = z\mathcal{H}(x)M(x,z), \quad M(0,z) = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}\right).$$

If $\mathcal{H} \notin L^1(\mathbb{R}_+)$ then the Weyl function $m_{\mathcal{H}}$ is well-defined by

$$m_{\mathcal{H}}(z) = \lim_{x \to +\infty} \frac{M_{22}(x,z)}{M_{21}(x,z)}, \quad z \in \mathbb{C}_+.$$

Operator of the canonical system

For $\tilde{X} \in \operatorname{dom} \mathcal{D}_{\mathcal{H}}$, we define

$$\mathcal{D}_{\mathcal{H}} : \tilde{X} \mapsto \tilde{Y},$$

where $\tilde{Y}$ is the solution of the equation

$$J\tilde{X}'(x) = \mathcal{H}(x)\tilde{Y}(x), \qquad x \geqslant 0,$$

$$J\tilde{X}(x) = J\tilde{X}(0) + \int_0^x \mathcal{H}\tilde{Y}(x_1)\, dx_1, \qquad x \geqslant 0.$$

The Hilbert space $L^2(\mathcal{H})$ is defined by

$$L^2(\mathcal{H}) = \Big\{ X : \|X\|^2_{L^2(\mathcal{H})} = \int_{\mathbb{R}_+} \langle \mathcal{H}(x)X(x), X(x)\rangle_{\mathbb{C}^2} < \infty \Big\}.$$

The domain $\operatorname{dom} \mathcal{D}_{\mathcal{H}}$ is defined by

$$\Big\{ \tilde{X} \in \mathrm{AC}_{\mathrm{loc}}(\mathbb{R}_+) \cap L^2(\mathcal{H}) : \big\langle \tilde{X}(0), \left(\begin{smallmatrix} \sin\alpha \\ \cos\alpha \end{smallmatrix}\right) \big\rangle_{\mathbb{C}^2} = 0,\ \mathcal{D}_{\mathcal{H}}\tilde{X} \in L^2(\mathcal{H}) \Big\}.$$

Reduction of Dirac system to canonical Hamiltonian system

The Dirac system and the canonical system

$$JN' + QN = zN, \qquad\qquad JM' = z\mathcal{H}M.$$

Define $N_q(x)$ as the solution of $JN_q'(x) + Q(x)N_q(x) = 0$ and let

$$\mathcal{H}_q(x) = N_q^*(x)N_q(x), \qquad x \geqslant 0.$$

Then the matrix $M(x,z) = N_q^{-1}(x)N(x,z)$ is the solution of the canonical system with the Hamiltonian $\mathcal{H}_q$.

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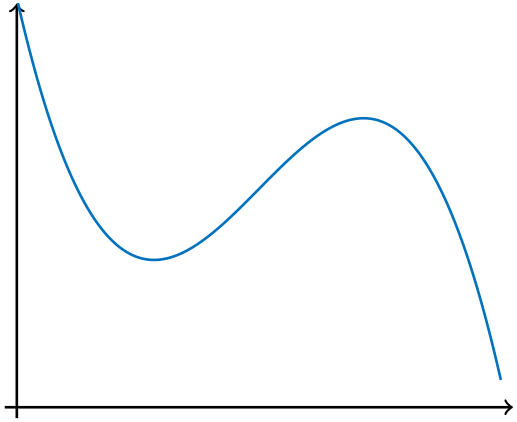
Theorem

Let $q \in L^2$. Then the Dirac operator $\mathcal{D}_q$ in $L^2(\mathbb{R}_+, \mathbb{C}^2)$ is unitarily equivalent to the operator $\mathcal{D}_{\mathcal{H}_q}$ in $L^2(\mathcal{H}_q)$ defined on the domain for the same boundary condition α. The unitary equivalence is given by the multiplication operator $U : X \mapsto N_q^{-1}X$ from $L^2(\mathbb{R}_+, \mathbb{C}^2)$ to $L^2(\mathcal{H}_q)$.

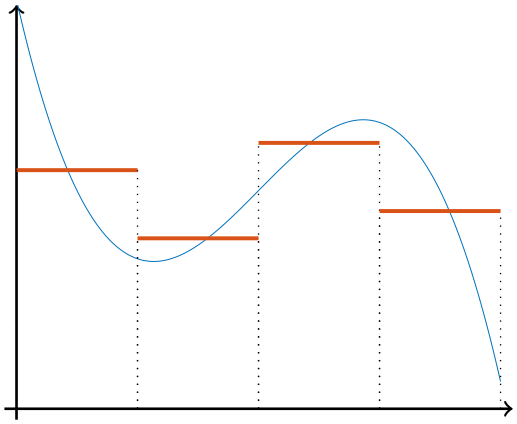
Discretization

Differential equations $JN' + QN = zN$, $JN_q' + QN_q = 0$, $JM' = z\mathcal{H}_q M$ with $\mathcal{H}_q = N_q^* N_q$.

Potential q



Averaged potential q_ℓ



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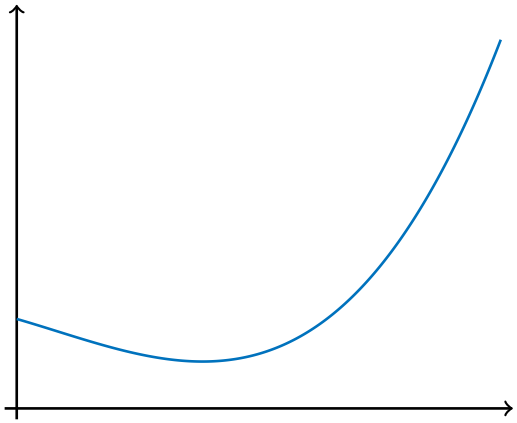
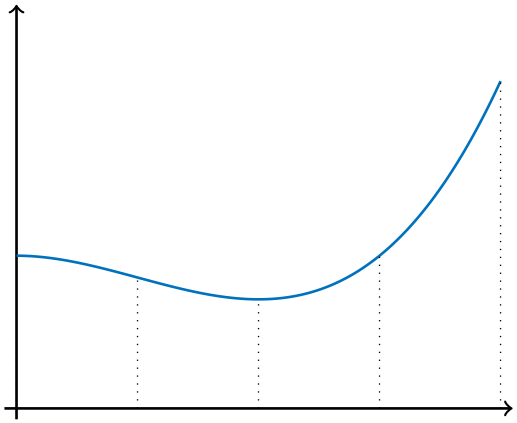
Hamiltonian $\mathcal{H}_q$



Hamiltonian $\mathcal{H}_{q_\ell}$



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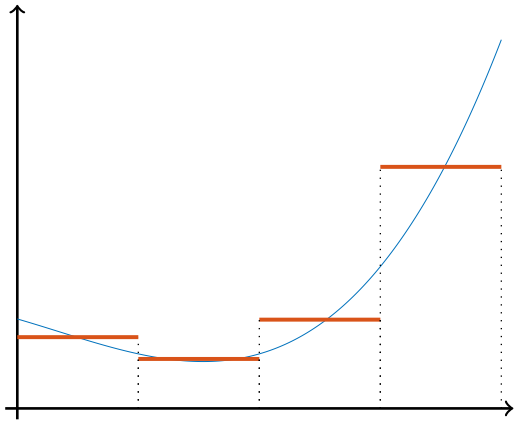

Measure-valued potential

Let q be a complex Borel measure with locally finite variation $|q|$. It is possible to define a self-adjoint Dirac operator $\mathcal{D}_q\colon X \mapsto Y$, where

$$JX'(x) + QX(x) = Y(x), \qquad x \geqslant 0.$$

$$JX(x) + \int_{[0,x]} g(Q(\{x_1\})J^*)Q(x_1)X(x_1) = JX(0-) + \int_{[0,x]} Y(x_1)\,dx_1.$$

Reduction to canonical system is done similarly. Define the solution N_q of

$$JN_q'(x) + Q(x)N_q(x) = 0, \qquad \lim_{x\to 0,\, x<0} N_q(x) = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}\right),$$

introduce the canonical system with the Hamiltonian

$$\mathcal{H}_q(x) = N_q^*(x)N_q(x)$$

and prove the theorem on unitary equivalence.

Relativistic Kronig-Penney model

We have $\operatorname{supp} q \subset \ell\mathbb{Z}_+$ for some $\ell > 0$, $q = \sum_{k \geqslant 0} q_k \delta_{\ell k}$.

The solution of $JN_q'(x) + Q(x)N_q(x) = 0$ satisfies

$$N_q(x) = e^{JQ(\{x\})} N_q(x-).$$

For arbitrary $x \geqslant 0$ we have

$$N_q(x) = e^{JQ_n} e^{JQ_{n-1}} \cdot \ldots \cdot e^{JQ_0}, \quad n = \lfloor x/\ell \rfloor.$$

Weyl functions of the piece-wise constant Hamiltonians

Lemma

The Weyl function of $\mathcal{H}_q$ is π/ℓ-periodic.

Proof.

$$N_q(x) = e^{JQ_n} e^{JQ_{n-1}} \cdot \ldots \cdot e^{JQ_0}, \qquad \mathcal{H}_q = N_q^* N_q,$$

$$JM_q' = z\mathcal{H}_q M_q, \qquad m_{\mathcal{H}}(z) = \lim_{x\to+\infty} \frac{M_{22}(x,z)}{M_{21}(x,z)},$$

$$M_q(\ell n, z) = M_{n-1}(z) M_{n-2}(z) \cdots M_0(z),$$

$$M_k(z) = \sqrt{\mathcal{H}(\ell k)}^{-1} \begin{pmatrix} \cos \ell z & \sin \ell z \\ -\sin \ell z & \cos \ell z \end{pmatrix} \sqrt{\mathcal{H}(\ell k)}.$$

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□

In particular, $F(e^{2i\ell z}) = f(z)$, $F\colon \mathbb{D} \to \mathbb{D}$ is well-defined.

Weyl functions of the piece-wise constant Hamiltonians II

Consider the bijective functions $\varkappa\colon \mathbb{D} \to \mathbb{C}$ defined by

$$\varkappa(w) = \frac{\overline{w}}{2|w|} \log \frac{1+|w|}{1-|w|}, \qquad \varkappa^{-1}(z) = \frac{\overline{z}}{|z|} \tanh |z|.$$

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Theorem (R. Bessonov, P.G., 2025)

For $q = \sum_{k \geqslant 0} q_k \delta_{\ell k}$ *we have* $q_0 = \varkappa(f_q(\infty))$, *where* $f_q(\infty) = \lim_{y \to \infty} f_q(iy)$.

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Theorem (R. Bessonov, P.G., 2025)

The Schur functions f_q *and* f_{q_ℓ} *of* $q = \sum_{k \geqslant 0} q_k \delta_{\ell k}$ *and* $q_\ell = \sum_{k \geqslant 0} q_{k+1} \delta_{\ell k}$ *are related by the*

$$e^{2i\ell z} f_{q_\ell}(z) = \frac{f_q(z) - f_q(\infty)}{1 - \overline{f_q(\infty)} f_q(z)}, \qquad z \in \mathbb{C}_+.$$

Schur's algorithm

For $F_q(e^{2i\ell z}) = f_q(z)$ and $F_{q_\ell}(e^{2i\ell z}) = f_{q_\ell}(z)$ we have

$$e^{2i\ell z} f_{q_\ell}(z) = \frac{f_q(z) - f_q(\infty)}{1 - \overline{f_q(\infty)} f_q(z)}, \qquad zF_{q_\ell}(z) = \frac{F_q(z) - F_q(0)}{1 - \overline{F_q(0)} F_q(z)}.$$

The classical Schur's algorithm in $\mathbb{D}$ constructs $F_0, F_1, \ldots$ as

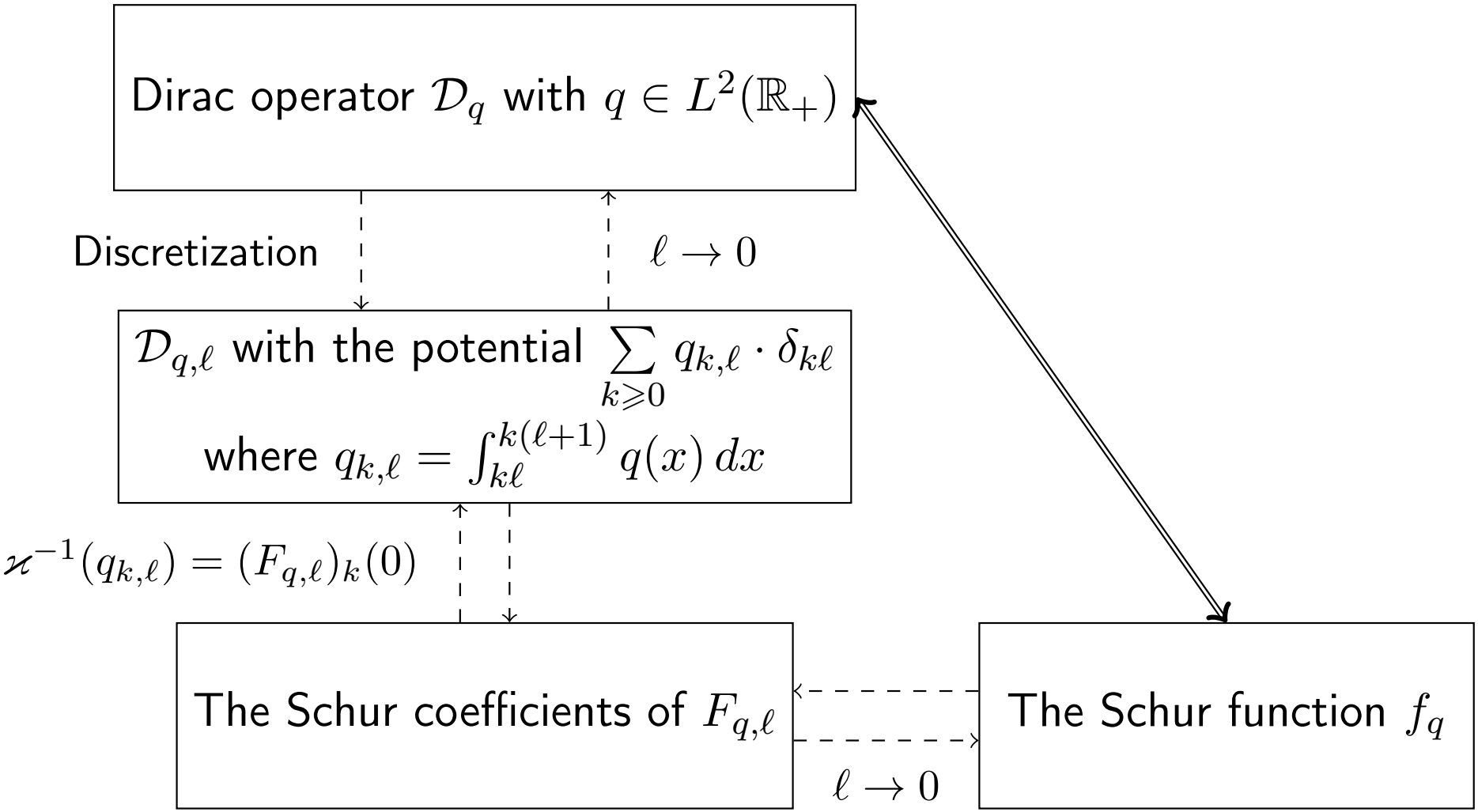
$$F_0 = F, \qquad zF_{k+1} = \frac{F_k - F_k(0)}{1 - \overline{F_k(0)} F_k}, \qquad k \geqslant 0.$$

The sequence $\{F_n(0)\}_{n \geqslant 0}$ is called the Schur coefficients of F.

Theorem (Verblunsky's theorem)

The mapping $F \mapsto \{F_n(0)\}_{n \geqslant 0}$ is a bijection from the set of Schur functions to $\mathbb{D}^{\times \infty}$.

Discretization + reduction



Stability result for Schur's algorithm

Theorem (R. Bessonov, P.G., 2025)

For Schur functions F, G and $r \in (0,1)$ we have

$$\sum_{k=0}^{\infty} r^k |\hat{F}(k) - \hat{G}(k)| \approx \sum_{k=0}^{\infty} r^k |F_k(0) - G_k(0)|,$$

if $1 - r \geqslant 12\psi(F)$, $1 - r \geqslant 12\psi(G)$, where

$$\psi(F) = -\int_{\mathbb{T}} \log(1 - |F|^2)\, dm, \qquad \psi(G) = -\int_{\mathbb{T}} \log(1 - |G|^2)\, dm.$$

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Theorem (R. Bessonov, P.G., 2025)

Let q, $\tilde{q} \in L^2(\mathbb{R}_+)$, and let $f_q, f_{\tilde{q}}$ be the Schur functions of the corresponding Dirac operators $\mathcal{D}_q$, $\mathcal{D}_{\tilde{q}}$. Then we have

$$\int_{\mathbb{R}_+} |\hat{f}_q(\xi) - \hat{f}_{\tilde{q}}(\xi)| e^{-A\xi}\, d\xi \approx \int_{\mathbb{R}_+} |q(\xi) - \tilde{q}(\xi)| e^{-2A\xi}\, d\xi$$

for any $A \geqslant 12 \max\left(\|q\|^2_{L^2(\mathbb{R}_+)}, \|\tilde{q}\|^2_{L^2(\mathbb{R}_+)}\right)$.

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